Problem 1 Cyclic Rotation Python Solution

```
def solution(A, K):
    # Get the length of the array
    N = len(A)

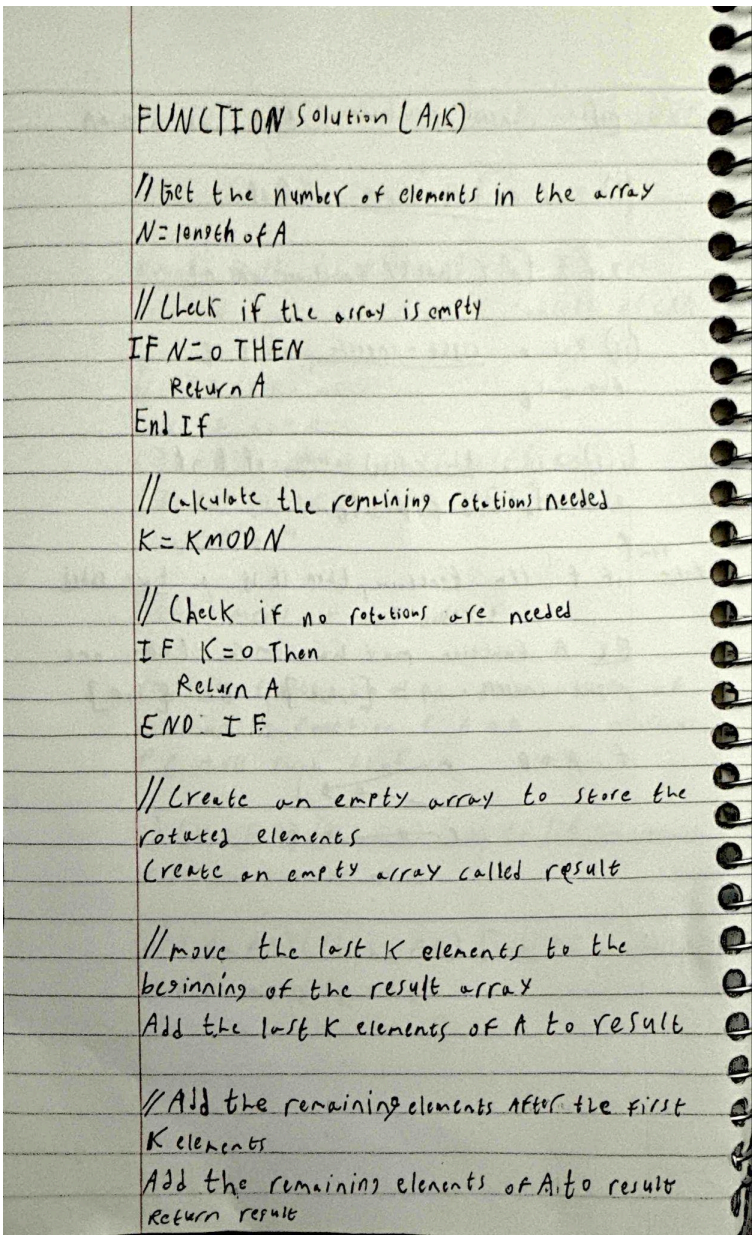
    # Check if the array is empty
    if N == 0:
        return A

    # Calculate the remaining rotations needed
    K = K % N

    # Check if no rotations are needed
    if K == 0:
        return A

    # Move the last K elements to the beginning
    # Keep the remaining elements at the end
    return A[-K:] + A[:-K]
```

Problem 1 pseudocode solution



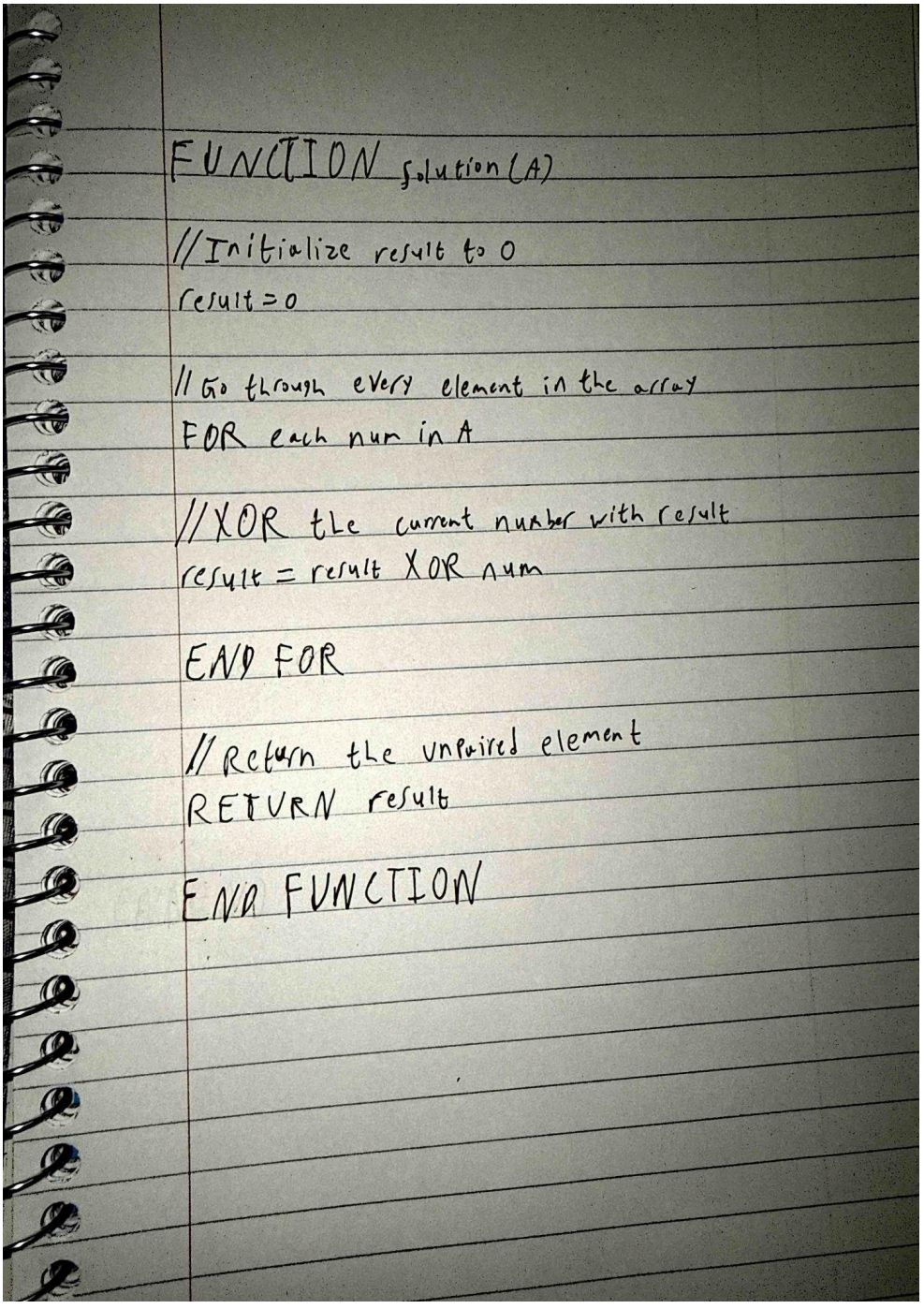
Problem 2 Odd Occurrences In Array Python Solution

```python
def solution(A):
    # Initialize the result to 0
    result = 0

    # Go through every element in the array
    for num in A:
        # XOR the current number with the result
        result = result ^ num

    # Return the unpaired element
    return result
```

Problem 2 pseudocode solution:



```
FUNCTION solution(A)

//Initialize result to 0
result = 0

//Go through every element in the array
FOR each num in A

//XOR the current number with result
result = result XOR num

END FOR

//Return the unpaired element
RETURN result

END FUNCTION
```

Problem 3 FrogJmp python solution

```python
def solution(X, Y, D):
    # Calculate the distance the frog needs to travel
    distance = Y - X

    # Calculate the number of complete jumps
    jumps = distance // D

    # Check if there is any remaining distance
    if distance % D != 0:
        jumps += 1

    # Return the minimum number of jumps
    return jumps
```

Problem 3 Pseudocode solution

```
FUNCTION Solution(X, Y, D)

// Calculate the distance to the target
distance = Y - X

// Calculate the num of complete jumps
jumps = distance DIV D

// Check if there is any remaining distance
IF distance MOD D != 0 THEN
        jumps = jumps + 1
END IF

// Return the min num of jumps
RETURN jumps
END FUNCTION
```

Problem 4 Passing Cars python solution

```python
def solution(A):
    # Initialize the number of eastbound cars
    east_cars = 0

    # Initialize the total number of passing cars
    passing_cars = 0

    # Go through every car in the array
    for car in A:

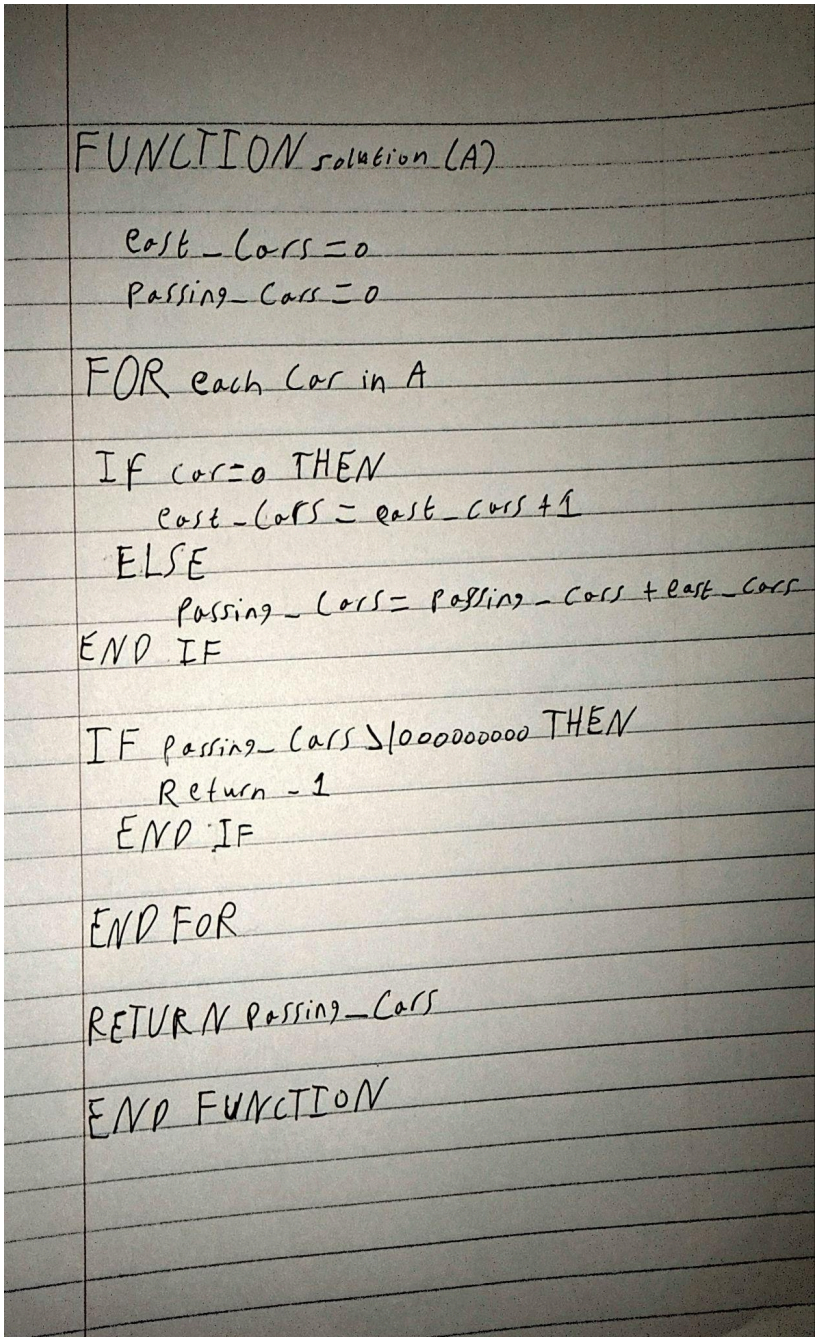
        # Check if the car is traveling east
        if car == 0:
            east_cars += 1

        # Otherwise, the car is traveling west
        else:
            # Add all previous eastbound cars to the passing count
            passing_cars += east_cars

        # Check if the passing count exceeds one billion
        if passing_cars > 1000000000:
            return -1

    # Return the total number of passing cars
    return passing_cars
```

Problem 4 pseudocode solution



```
FUNCTION solution (A)

    east_cars = 0
    passing_cars = 0

FOR each car in A

  IF car=0 THEN
      east_cars = east_cars + 1
   ELSE
      passing_cars = passing_cars + east_cars
END IF

IF passing_cars > 1000000000 THEN
     Return -1
  END IF

END FOR

RETURN passing_cars

END FUNCTION
```

Problem 5 Missing Integer python solution

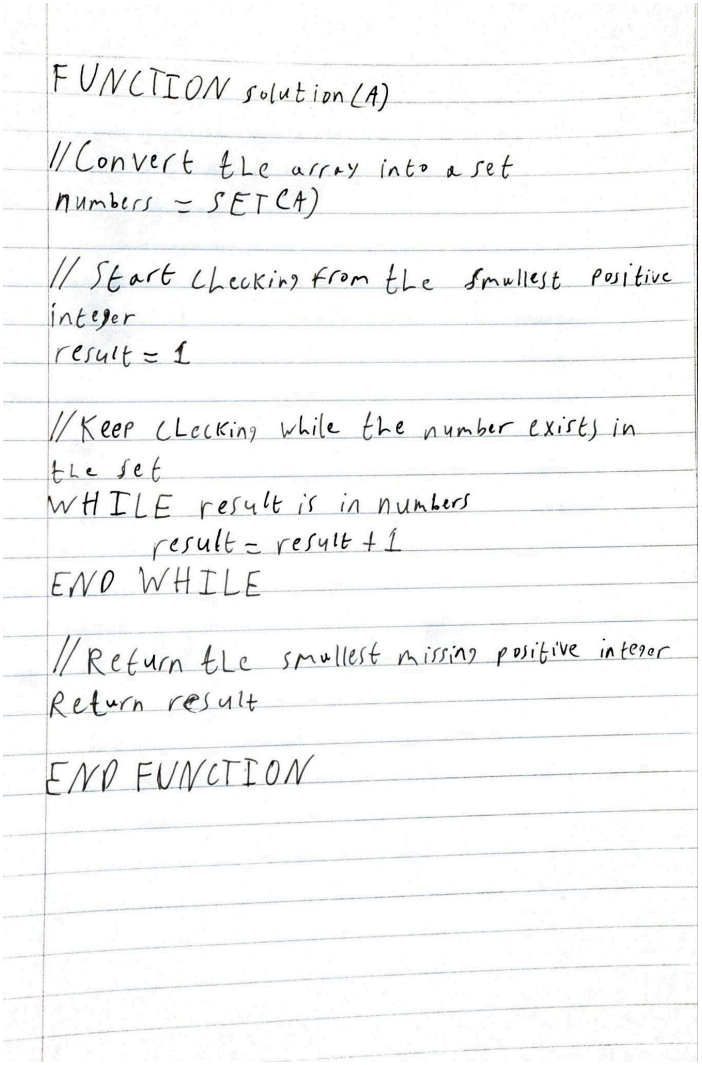
```python
def solution(A):
    # Convert the array into a set to make searching faster
    numbers = set(A)

    # Start checking from the smallest positive integer
    result = 1

    # Keep checking while the number exists in the array
    while result in numbers:
        result += 1

    # Return the smallest positive integer that is missing
    return result
```

Problem 5 pseudocode solution



```
FUNCTION solution(A)

// Convert the array into a set
numbers = SET(A)

// Start checking from the smallest positive integer
result = 1

// Keep checking while the number exists in the set
WHILE result is in numbers
    result = result + 1
END WHILE

// Return the smallest missing positive integer
Return result

END FUNCTION
```

Sources:

https://chatgpt.com/share/6af13a28-0214-83ea-95b8-ad4bd54c45b1